Tsz Fei Wang

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Education

University of Waterloo | Honours Computer Science (Co-op) — Sept 2026 – May 2031 (expected)

Skills & Technologies

Languages: Python, Java, C, C++, C#, TypeScript, JavaScript, HTML/CSS, SQL, Bash, GLSL
Frameworks & Libraries: React, Next.js, Node.js, FastAPI, Three.js, PyTorch, NumPy, Pandas, Qt/QML
Tools & Platforms: Git, Linux, Docker, GitHub Actions, Firebase, Elasticsearch, Vercel, Cloudflare Workers, Postman, Jupyter, KiCad

Projects

huddle | *Node.js, Cloudflare Workers, MCP* — *Oct 2026 – Present*

- Designed a coordination layer that lets multiple **Claude Code** agents on separate machines edit one repository at once, using **per-file locks** that make other checkouts read-only and stream edits to them live.
- Implemented a dependency-free **Node.js** relay server deployed on **Cloudflare Workers**, with agent-to-agent chat, a web dashboard and git history sync between checkouts.

Submerge | *GLSL, QML, Python* — *Oct 2026 – Present*

- Developed an animated koi theme for KDE Plasma 5 and 6 in **GLSL** and **QML**, with a water ripple simulation, a live system HUD and **power-aware frame pacing**, reaching **~1,290 downloads** on the KDE Store.
- Resolved blank rendering on **NVIDIA/Wayland** after a user report by porting the shaders to GLSL ES.

Physical Studio | *Python, React, Three.js, Docker* | *Winner, Hack the Valley 11* — *Oct 2026*

- Engineered the **MCP server** that lets an LLM design a hardware project from a library of measured parts and check its own design for collisions and clearance.
- Implemented GLB scene export to a **React Three Fiber** viewer that animates each step of the generated assembly guide.
- Deployed the app with **Docker Compose** and Caddy on a Vultr server at physicalstudio.work as part of a team of three.

Technova Hackathon Badge | *KiCad, Hardware Lead* — *Oct 2026*

- Led hardware design of an **NFC event badge** for **~300 attendees** in **KiCad**, with a RISC-V MCU, LED ring and USB-C charging.
- Routed the double-sided board (**1,300+ tracks**) to a clean DRC using Freerouting with manual fixes.
- Prepared the **JLCPCB** manufacturing package (gerbers, LCSC-matched BOM, placement file) and modelled the per-badge cost of each option.

Experience

Century Frontier Asset Management — Jul 2026 – Aug 2026
Quantitative Research Intern — *Shenzhen, CN*

- Designed and implemented a **point-in-time backtesting pipeline** (**Python**, **pandas**, yfinance), eliminating lookahead bias in **LLM-driven** equity research.
- Built a **model-agnostic evaluation framework** benchmarking stock-preference and inter-stock-relationship predictions across **9 LLM architectures** and **104 tickers**, isolating true signal quality from raw model capability.

Krembil Research Institute, World Community Grid — Mar 2026 – Jun 2026
Software Engineer Intern — *Toronto, ON*

- Built a **full-stack web data access portal** for Catrin (Catalogue of Transcriptional Regulatory Interactions), unifying **~4 million** TF-TG associations across **33,000 genes** from **9** independent data sources into a single queryable interface.

Mackenzie Cybersecurity Team — Jun 2025 – Jun 2026
Team Captain, President — *Toronto, ON*

- Designed and delivered **cybersecurity** curriculum and class presentations reaching **60+ students** to raise digital-safety awareness school-wide.
- **Captained team to a national #1 finish** in CyberTitan, Canada's largest **cyberdefense** competition, coordinating strategy and technical training across the roster.

Non-Trivial Research Fellowship — Mar 2025 – Aug 2025
Machine Learning Research Fellow — *Remote*

- Built an **audio deepfake detection** model in **Python** on **Kaggle**, training and evaluating a classifier that distinguishes real from AI-synthesized speech.

Off Script, Waterloo Tech Week | *Founder & Host, 120+ registrations* — Sept 2026

Awards & Honors

Gold & Silver Medalist, IT Systems Networking, Skills Ontario (Regional & Provincial) — Mar 2026 – May 2026
National Champion x2, 1st in Canada, Cybersecurity, $1,000 scholarship — May 2025, Mar 2026
National Finalist, 2nd in Canada, Cybersecurity, $1,600 scholarship — Mar 2025
1st Place, Canada, Mirai Project, Technology Innovation in Healthcare, $400 award — Jan 2025